

Dijkstra Algorithm

Difficulty: Medium Accuracy: 50.83% Submissions: 317K+ Points: 4 Average Time: 25m

Given an undirected, weighted graph with V vertices numbered from 0 to $V-1$ and E edges, represented by 2d array `edges[][]`, where `edges[i]=[u, v, w]` represents the edge between the nodes u and v having w edge weight.

You have to find the shortest distance of all the vertices from the source vertex `src`, and return an array of integers where the i th element denotes the shortest distance between i th node and source vertex `src`.

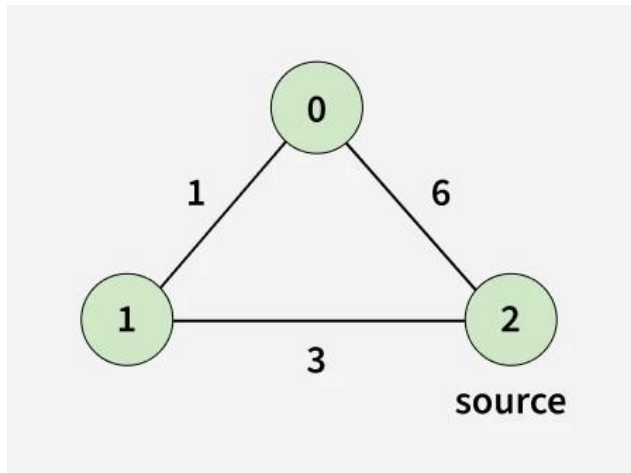
Note: The Graph is connected and doesn't contain any negative weight edge.
It is guaranteed that all the shortest distance will fit in a 32-bit integer.

Examples:

Input: $V = 3$, `edges[][] = [[0, 1, 1], [1, 2, 3], [0, 2, 6]]`, `src = 2`

Output: `[4, 3, 0]`

Explanation:



Shortest Paths:

For 2 to 0 minimum distance will be 4. By following path $2 \rightarrow 1 \rightarrow 0$

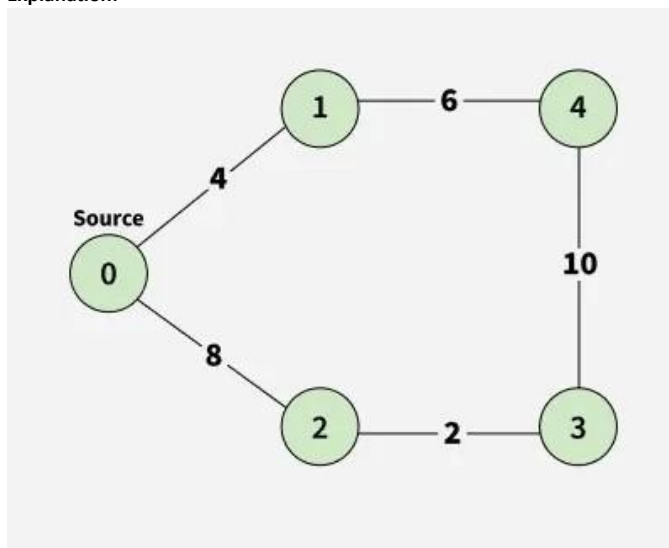
For 2 to 1 minimum distance will be 3. By following path $2 \rightarrow 1$

For 2 to 2 minimum distance will be 0. By following path $2 \rightarrow 2$

Input: $V = 5$, `edges[][] = [[0, 1, 4], [0, 2, 8], [1, 4, 6], [2, 3, 2], [3, 4, 10]]`, `src = 0`

Output: `[0, 4, 8, 10, 10]`

Explanation:



Shortest Paths:

For 0 to 1 minimum distance will be 4. By following path $0 \rightarrow 1$

For 0 to 2 minimum distance will be 8. By following path $0 \rightarrow 2$

For 0 to 3 minimum distance will be 10. By following path $0 \rightarrow 2 \rightarrow 3$

For 0 to 4 minimum distance will be 10. By following path $0 \rightarrow 1 \rightarrow 4$